

Smart Farm Field Intelligence Platform

1. Project Overview

The Smart Farm Field Intelligence Platform is a digital agriculture system designed to help farmers and agronomists make data-driven decisions about their fields. Users draw a farm boundary on an interactive map, and the system analyzes that polygon using weather data, soil intelligence, satellite imagery, and agronomic models.

The platform generates a comprehensive field intelligence report including crop suitability, irrigation recommendations, soil insights, crop health monitoring, fertilizer planning, pest control recommendations, profitability analysis, and nearby Nutrien branch information.

This system integrates multiple environmental datasets and transforms them into actionable agricultural insights through a centralized field dashboard.

2. Objectives

Primary objectives of the platform:

- Enable farmers to digitally map farm boundaries.
 - Analyze soil and environmental conditions of the field.
 - Recommend suitable crops and farming practices.
 - Provide irrigation and fertilizer planning guidance.
 - Monitor crop health using satellite imagery.
 - Generate pest and disease spray recommendations.
 - Estimate crop profitability before planting.
 - Help farmers locate nearby Nutrien branches and pricing.
 - Provide alerts for environmental risks such as drought or frost.
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3. System Scope

The system provides the following capabilities:

- Interactive farm boundary mapping
- Environmental data aggregation
- Satellite crop monitoring
- Agronomic decision support
- Fertilizer and pesticide recommendations

- Profitability estimation
- Nutrien retail branch discovery
- Farm analytics dashboard

The platform will initially be implemented as a **web application**, with potential expansion to mobile platforms.

Phase 1 – Discover (Problem Understanding)

Problem Statement

Farmers often lack localized insights about soil health, weather trends, crop suitability, irrigation needs, and profitability. Agricultural decisions are frequently based on experience rather than integrated environmental intelligence.

This project aims to build a unified digital platform that combines multiple geospatial datasets and AI models to provide precise farm-level recommendations.

Stakeholders

Primary stakeholders:

- Farmers
 - Agronomists
 - Agricultural researchers
 - Nutrien retail network
 - AgTech companies
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Key User Personas

Farmer

Needs crop recommendations, fertilizer advice, irrigation scheduling, and pest alerts.

Agronomist

Analyzes soil trends, crop performance, and field health.

Retail Network (Nutrien)

Provides fertilizers, pesticides, and agronomy services to farmers.

Core User Journey

1. User opens the platform.
 2. User draws the farm boundary on the map.
 3. Field polygon is saved.
 4. User clicks **Analyze Field**.
 5. System processes environmental and satellite data.
 6. Field Intelligence Dashboard displays results.
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Phase 2 – Design (System Architecture)

High-Level Architecture

The system consists of the following layers:

Frontend Layer

Interactive web application for visualization and user interaction.

Backend API Layer

Processes geospatial data, integrates external APIs, and runs agronomic models.

Data Integration Layer

Connects external datasets including weather, soil, and satellite imagery.

Analytics Layer

Processes environmental data to generate agricultural recommendations.

Storage Layer

Stores field polygons, user data, and historical analytics.

System Components

Frontend - React.js - Leaflet or Mapbox - Chart.js / Recharts

Backend - Node.js

Database - PostgreSQL - PostGIS extension

Geospatial Tools - GeoJSON - Turf.js

External APIs - SoilGrids - Open-Meteo - NASA POWER - Sentinel-2 - Sentinel Hub / Google Earth Engine

Phase 3 – Develop (Module Architecture)

Module 1 – Field Mapping

Purpose

Allow users to draw and store farm boundaries.

Features - Interactive map - Polygon drawing tool - Area calculation - GeoJSON storage

Output Data

Field ID

Field Name

Coordinates

Field Area

Module 2 – Weather Intelligence

Provides localized weather analytics for the farm.

Data Collected - Current weather - 7–14 day forecast - Rainfall prediction - Temperature - Humidity - Wind speed

Data Sources - Open-Meteo - NASA POWER - OpenWeather

Visualizations - Temperature trend charts - Rainfall timeline - Wind indicators

Example Output

Temperature: 28°C

Rainfall forecast: 45 mm next 7 days

Module 3 – Soil Intelligence

Retrieves soil properties for the farm polygon.

Parameters - Soil pH - Organic carbon - Nitrogen estimation - Clay percentage - Sand percentage - Soil texture classification

Data Source SoilGrids

Example Output

Soil Type: Loam

pH: 6.4

Organic Carbon: 1.1%

Module 4 – Historical Soil Intelligence

Tracks soil health trends across multiple years.

Features - Store seasonal soil test data - Analyze nutrient depletion - Track organic carbon trends - Monitor soil acidity changes

Example Output

Soil pH Trend

2023 – 6.7

2024 – 6.5

2025 – 6.3

System Alert

Soil acidity increasing.

Module 5 – Crop Suitability Engine

Evaluates which crops are best suited for the field.

Inputs - Soil properties - Climate conditions - Rainfall patterns

Scoring Model

Suitability Score = (Soil Compatibility × 40%) + (Climate Compatibility × 35%) + (Rainfall Suitability × 25%)

Example Output

Wheat – 90%

Maize – 82%

Soybean – 75%

Module 6 – Irrigation Recommendation Engine

Calculates irrigation schedules based on soil moisture and weather.

Inputs - Soil water holding capacity - Evapotranspiration - Rainfall forecast

Output Example

Next irrigation: 3 days

Water required: 12 mm

Module 7 – Satellite Crop Health Monitoring

Uses satellite imagery to analyze vegetation health.

Satellite Source Sentinel-2 imagery

Vegetation Index NDVI (Normalized Difference Vegetation Index)

NDVI Formula

$$\text{NDVI} = (\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red})$$

Outputs - NDVI heatmap - Crop stress detection - Vegetation health score

Tools - Sentinel Hub - Google Earth Engine - OpenEO

Module 8 – Fertilizer Recommendation Engine (NPK Calculator)

Determines required fertilizer quantities.

Inputs - Soil nutrient levels - Target crop - Field area

Calculation

Required Nutrient = Crop Requirement – Soil Availability

Example Output

Crop: Maize

Nitrogen Required: 120 kg/ha

Phosphorus Required: 60 kg/ha

Potassium Required: 40 kg/ha

Recommended Fertilizer

Urea: 260 kg/ha

DAP: 130 kg/ha

MOP: 65 kg/ha

Module 9 – Pesticide Spray Recommendation

Provides crop protection guidance.

Inputs - Crop type - Growth stage - Weather conditions - Pest risk indicators

Outputs - Recommended pesticide - Spray dosage - Spray timing - Safety instructions

Example

Crop: Soybean
Pest Risk: Aphids

Recommendation

Product: Lambda-cyhalothrin
Dosage: 250 ml per acre

Module 10 – Risk & Alert Engine

Detects environmental risks affecting crop growth.

Risk Types - Drought - Frost - Heat stress - Heavy rainfall - Flooding

Example Alert

Heat stress expected within next 48 hours.

Module 11 – Field Profitability Analysis

Estimates potential financial outcomes for crops.

Inputs - Crop type - Expected yield - Market price - Input costs

Calculation

Profit = Revenue – Total Input Cost

Example

Yield: 3.2 tons/acre
Price: \$220/ton

Revenue: \$704
Costs: \$410

Estimated Profit: \$294 per acre

Module 12 – Nutrien Branch Locator

Helps farmers locate nearby Nutrien retail branches.

Features - Branch map visualization - Distance from farm - Contact information -
Navigation directions

Example Output

Nearest Branch
Nutrien Saskatoon

Distance: 12.4 km

Services - Fertilizers - Crop Protection - Agronomy Support

Module 13 – Nutrien Branch Price Comparison

Helps farmers find the most cost-effective supplier.

Features - Compare fertilizer prices across branches - Identify seasonal discounts -
Recommend lowest price

Example

Urea (50kg)

Branch A – \$32

Branch B – \$29 (Best Price)

Branch C – \$31

Module 14 – Field Intelligence Dashboard

Central interface displaying all insights.

Dashboard Components

- Weather summary
- Soil summary
- Crop suitability ranking
- NDVI vegetation map
- Fertilizer recommendations
- Irrigation schedule
- Pest spray suggestions
- Profitability analysis
- Risk alerts
- Nearby Nutrien branches

This dashboard acts as the **digital health report for the farm.**

Phase 5 – Monitor (Operations)

Monitoring Metrics

- API response time
 - NDVI processing time
 - Data accuracy
 - User activity
 - Error rates
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Logging

Centralized logging using:

- ELK Stack
 - Datadog
 - CloudWatch
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System Alerts

Operational alerts triggered for:

- API failures
 - Satellite processing errors
 - External dataset downtime
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Security Considerations

- Secure API key storage
 - HTTPS encryption
 - Authentication for users
 - Input validation for geospatial data
 - Rate limiting for APIs
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Testing Strategy

Unit Testing

Test individual modules such as NPK calculations and crop suitability scoring.

Integration Testing

Verify data flow between external APIs and backend services.

Geospatial Testing

Validate polygon analysis accuracy.

User Acceptance Testing

Confirm dashboard recommendations are correct.

Future Enhancements

- AI disease detection from crop images
 - Drone-based crop monitoring
 - Yield prediction models using machine learning
 - Mobile app for farmers
 - Voice-based farm assistant
 - Integration with farm machinery sensors
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Example Field Analysis Report

Field: North Farm Plot

Weather

Rainfall forecast: 45 mm next 7 days

Soil

pH: 6.4

Texture: Loam

Crop Suitability

Wheat – 92%

Maize – 84%

Fertilizer Plan

Urea: 260 kg/ha

DAP: 130 kg/ha

NDVI

Healthy vegetation detected

Irrigation

Next irrigation recommended in 4 days

Profitability

Estimated profit: \$294 per acre

Nearby Nutrien Branch
Distance: 12.4 km

Conclusion

The Smart Farm Field Intelligence Platform integrates geospatial mapping, environmental datasets, and agronomic models to deliver comprehensive farm insights. By combining weather intelligence, soil analysis, satellite monitoring, and retail integration, the system provides a complete digital decision support platform for modern agriculture.